Lecture 29 : Computation of Project Characteristics Using PERT/CPM

Introduction

In the dynamic world of project management, effectively scheduling and controlling tasks is paramount to success. This lecture delves into the foundational techniques of Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM), which are indispensable tools for project managers. While both methods utilize a task network to visualize activities and their dependencies, they differ fundamentally in their approach to task durations. CPM employs deterministic task times, assuming a fixed duration for each activity, which simplifies calculations but may not reflect real-world uncertainties. Conversely, PERT acknowledges the inherent unpredictability of tasks by using probabilistic task times, offering a more realistic estimation for projects where completion times are uncertain.

The primary objective of PERT/CPM is to enable project managers to determine various critical project characteristics. These characteristics, once identified, form the basis for informed decision-making, resource allocation, and proactive risk management throughout the project lifecycle. By understanding the earliest and latest possible start and finish times for each activity, as well as the flexibility (or lack thereof) within the schedule, managers can strategically deploy resources, mitigate potential delays, and ensure the project stays on track. This lecture will first explore the simpler CPM approach to calculating these characteristics before touching upon the statistical considerations of PERT, providing a comprehensive understanding of how these powerful techniques empower effective project scheduling and control.

Core Concepts

Term	Definition	Significance
PERT (Program Evaluation and Review Technique)	A project management technique that uses probabilistic task times to account for uncertainty in activity durations, making it suitable for projects with unpredictable tasks.	Provides a more realistic schedule for complex projects by incorporating statistical estimates for task completion, aiding in risk assessment.

Term	Definition	Significance
CPM (Critical Path Method)	A project management technique that uses deterministic task times, assuming fixed durations for activities, making it simpler for projects with well-defined tasks.	Offers a straightforward way to calculate project duration and identify critical activities when task times are known and stable.
Earliest Start Time (ES)	The earliest possible moment an activity can begin, considering the completion of all its preceding activities.	Crucial for planning when resources can be deployed and when subsequent activities can commence without delay.
Earliest Finish Time (EF)	The earliest possible moment an activity can be completed, calculated by adding its duration to its earliest start time.	Helps in forecasting the earliest possible completion of individual tasks and the overall project.
Latest Start Time (LS)	The latest possible moment an activity can begin without delaying the overall project completion date.	Provides flexibility to the project manager; starting later than this time will inevitably delay the project.
Latest Finish Time (LF)	The latest possible moment an activity can be completed without delaying the overall project completion date.	Defines the absolute deadline for an activity's completion to maintain the project schedule.
Slack Time (Float)	The amount of time an activity can be delayed without affecting the earliest start time of any successor activity or the overall project completion date.	Represents the flexibility available for an activity, allowing managers to reallocate resources or manage unexpected delays without impacting the project timeline.
Critical Path		

Term	Definition	Significance
	The longest sequence of activities in a project network, where all activities have zero slack, determining the minimum possible project duration.	Identifies the most crucial activities that must be completed on time to avoid project delays, requiring close monitoring and control.
Burst Point	A node in the task network where a single activity has multiple successor activities, meaning its completion allows several subsequent tasks to begin.	Highlights dependencies where one task's completion unlocks multiple parallel paths, impacting resource allocation for subsequent tasks.
Merge Point	A node in the task network where an activity has multiple predecessor activities, meaning it can only start after all its preceding tasks are completed.	Indicates a point where multiple paths converge, requiring careful coordination to ensure all prerequisites are met before the next task can commence.

Detailed Analysis

1. Understanding Project Activity Attributes

Project scheduling using PERT/CPM involves identifying and quantifying several attributes for each activity. Beyond the inherent duration of a task, which is an initial estimate (e.g., 7 days for "write software documentation"), the network analysis allows for the computation of dynamic attributes crucial for project management.

- **Activity Duration**: This is the estimated time required to complete a specific task. It's typically expressed in days or weeks for high-level scheduling.
- **Earliest Start Time (ES)**: This is the earliest point in time an activity can begin. It's determined by the completion of all its preceding activities.
- **Earliest Finish Time (EF)**: This is the earliest point in time an activity can be completed. It's calculated by adding the activity's duration to its earliest start time (EF = ES + Duration).

- **Latest Start Time (LS)**: This is the latest point in time an activity can begin without causing a delay to the overall project completion date. If an activity starts later than its LS, the project will be delayed.
- **Latest Finish Time (LF)**: This is the latest point in time an activity can be completed without delaying the overall project completion date. If an activity finishes later than its LF, the project will be delayed. (LF = LS + Duration).
- **Slack Time (Float)**: This represents the amount of flexibility available for an activity. It's the difference between the latest start time and the earliest start time (Slack = LS - ES), which is also equal to the difference between the latest finish time and the earliest finish time (Slack = LF - EF). A positive slack indicates that an activity can be delayed without impacting the project timeline, offering the project manager some liberty in resource allocation. A zero slack indicates a critical activity.

2. PERT/CPM Terminology and Network Structure

The PERT/CPM methodology relies on a structured task network to represent activities and their interdependencies. To ensure the computational methods are straightforward, certain conventions are followed:

- **Single Start Node**: The network must have a single starting point. If multiple activities can begin independently, a dummy start node is introduced to consolidate them into a single origin.
- **Single Finish Node**: Similarly, the network must conclude with a single finish point. If multiple activities can end the project, a dummy finish node is used to merge them into a single termination.
- **Burst Point**: Occurs when a single activity's completion enables multiple subsequent activities to begin. For example, if activity 'B' completes, both 'C' and 'D' can start, making 'B' a burst point.
- **Merge Point**: Occurs when an activity requires the completion of multiple preceding activities before it can start. For instance, if activity 'E' can only begin after both 'C' and 'D' are finished, then 'E' is a merge point. 'C' and 'D' are predecessors of 'E', and 'E' is a successor of 'C' and 'D'.
- **Task Notation**: Activities are typically represented as rectangles. The activity name is placed in the center, and its duration is noted. The computed attributes (ES, EF, LS, LF, Float) are then added around the rectangle as they are calculated.

3. Time Measurement and Scheduling Levels

PERT/CPM employs a simplified approach to time measurement for high-level scheduling:

- **Day 0 Convention**: The project is considered to start at "day 0," which signifies the end of day 0 or the beginning of day 1. This convention avoids ambiguity and simplifies calculations.
- **Relative Time Units**: Scheduling is done in relative time units (e.g., days, weeks) rather than specific calendar dates. This high-level approach intentionally disregards weekends, public holidays, and resource availability constraints.
- **High-Level vs. Detailed Scheduling**: PERT/CPM provides a high-level schedule, focusing on the logical sequence and overall duration. More detailed scheduling, which incorporates calendar dates, holidays, and resource allocation, is typically performed using tools like Gantt charts after the PERT/CPM analysis is complete. This two-tiered approach allows for a simpler initial plan that is later refined with real-world constraints.

4. The Importance of Slack Time and Critical Path

Two of the most valuable outputs of PERT/CPM analysis for a project manager are the slack time for each activity and the identification of the critical path.

- **Slack Time for Flexibility**: Slack time provides the project manager with crucial flexibility. Activities with positive slack can be delayed by that amount of time without impacting the project's overall completion date. This allows managers to reallocate resources from tasks with slack to those that are falling behind, or to manage unexpected issues without causing project-wide delays.
- **The Critical Path**: This is defined as the sequence of activities from the project's start to its finish where all activities have zero slack. Any delay in an activity on the critical path will directly result in a delay of the entire project.
- **Monitoring Critical Activities**: Project managers must pay special attention to critical activities. They are closely monitored, and corrective actions (e.g., deploying additional resources) are taken immediately if any critical activity shows signs of delay.
- **Minimum Project Duration**: The sum of the durations of all activities on the critical path represents the minimum possible time required to complete

the entire project. This provides a baseline for project duration and a target for efficient project execution.

5. The PERT/CPM Computation Technique

The computation of project characteristics using PERT/CPM involves a systematic three-step process:

- **Step 1: Forward Pass (Left-to-Right Calculation)**
- **Purpose**: To determine the Earliest Start (ES) and Earliest Finish (EF) times for all activities.
- **Process**: Starting from the project's initial activity (at day 0), the calculation moves forward through the network.
- **Rule for Single Predecessor**: For an activity with a single predecessor, its ES is equal to the EF of its predecessor.
- **Rule for Multiple Predecessors (Merge Point)**: For an activity with multiple predecessors (e.g., activity 'H' following 'E' and 'F'), its ES is set to the latest of the Earliest Finish times of all its immediate predecessors. This ensures that 'H' can only start once all its preceding tasks are completed.
- **Calculation**: EF = ES + Duration.
- **Step 2: Backward Pass (Right-to-Left Calculation)**
- **Purpose**: To determine the Latest Start (LS) and Latest Finish (LF) times for all activities.
- **Process**: Starting from the project's final activity (whose LF is typically set to the project's overall EF determined by the forward pass), the calculation moves backward through the network.
- **Rule for Single Successor**: For an activity with a single successor, its LF is equal to the LS of its successor.
- **Rule for Multiple Successors (Burst Point)**: For an activity with multiple successors (e.g., activity 'B' preceding 'D' and 'E'), its LF is set to the smallest of the Latest Start times of all its immediate successors. This ensures that 'B' completes in time for its earliest-starting critical successor.
- **Calculation**: LS = LF - Duration.
- **Step 3: Identification of Critical Path(s)**
- **Purpose**: To identify the sequence(s) of activities that have zero slack and thus determine the minimum project duration.

- **Process**: After both forward and backward passes are complete, the slack time (LS - ES or LF - EF) is calculated for every activity. Any activity with zero slack is a critical activity.
- **Marking**: The critical path(s) are then identified as the continuous sequence(s) of critical activities from the start to the finish of the project. These are often highlighted (e.g., in red) to draw the project manager's attention. There can be more than one critical path in a project.

Key Takeaways

- [ ] PERT uses probabilistic task times for uncertain projects, while CPM uses deterministic times for well-defined tasks.
- [ ] Project managers determine activity duration, earliest start/finish, latest start/finish, and slack time using PERT/CPM.
- [ ] Slack time (float) provides flexibility; it's the difference between latest and earliest start/finish times.
- [ ] The task network requires a single start and finish node, using dummy nodes if necessary.
- [ ] Burst points indicate one activity leading to multiple successors, while merge points indicate an activity requiring multiple predecessors.
- [ ] High-level PERT/CPM scheduling uses relative days (e.g., day 0) and does not account for calendar dates or holidays, which are handled in detailed Gantt charts.
- [ ] The critical path is the sequence of activities with zero slack, determining the minimum project duration.
- [ ] Any delay on a critical path activity will delay the entire project, requiring close monitoring.
- [ ] The forward pass calculates earliest start and finish times, taking the latest finish of predecessors at merge points.
- [ ] The backward pass calculates latest start and finish times, taking the smallest latest start of successors at burst points.

Conclusion

The computation of project characteristics using PERT/CPM provides project managers with an invaluable framework for planning, monitoring, and controlling complex endeavors. By systematically analyzing task dependencies and durations through forward and backward passes, managers gain a clear understanding of the project's timeline, identifying critical activities that demand immediate attention and areas where flexibility exists. The ability to pinpoint the critical path not only reveals

the shortest possible project duration but also highlights the tasks whose delays would directly impact the overall schedule, enabling proactive risk mitigation.

Furthermore, the concept of slack time empowers managers to optimize resource allocation, allowing for strategic shifts of manpower or equipment to address bottlenecks without jeopardizing the project's completion. While modern software tools automate these calculations, a fundamental understanding of the underlying PERT/CPM methodology remains essential for effective interpretation of results and informed decision-making. This foundational knowledge ensures that project managers can confidently navigate the complexities of project scheduling, ultimately leading to more efficient execution and successful project outcomes.